Lemma 1. *There is a smooth hyperelliptic curve $\Gamma_0/\mathbf{Q}$ of genus 4, a marked Weierstrass point $p_0 \in \Gamma_0$, and a tangential cover*

$$\pi_0 : (\Gamma_0, p_0) \longrightarrow (E_0, 0)$$

to an elliptic curve such that the complementary three-dimensional abelian quotient

$$A_0 = J(\Gamma_0)/\pi_0^* E_0$$

is absolutely simple over $\mathbf{C}$. Moreover, if

$$P_{\mathbf{Q}} = H^1(A_0, \mathbf{Q}) \hookrightarrow H^1(\Gamma_0, \mathbf{Q}),$$

then

$$F^1(P_{\mathbf{C}}) = H^0\big(\Gamma_0, K_{\Gamma_0}(-p_0)\big).$$

Proof. We use four external results, stated here in the precise forms needed.

(1) Robert S. Maier, *Lamé polynomials, hyperelliptic reductions and Lamé band structure*, paper_id=Maier2003, arXiv id=math-ph/0309005, theorem_id=definition (3.5), Table 3 ($\ell = 4$), Theorem L, and Section 6, proves this specialized complete statement. For the nonsingular curve $E_{g_2,g_3} : y^2 = 4x^3 - g_2 x - g_3$, with $4e_i^3 - g_2 e_i - g_3 = 0$, put

$$\mathsf{L}_4 = (B^3 - 52 g_2 B + 560 g_3) \prod_{i=1}^{3} (B^2 + 10 e_i B - 35 e_i^2 - 7 g_2).$$

If L_4 is square-free, then $\Gamma_4 : \nu^2 = \mathsf{L}_4$ is a smooth genus-four curve, and the rational functions in Theorem L define a degree-$4 \cdot 5/2 = 10$ algebraic spectral cover $\Gamma_4 \to E_{g_2,g_3}$, sending the spectral point at infinity to the origin.

(2) J. C. Eilbeck, V. Z. Enolskii, and E. Previato, *Varieties of elliptic solitons*, paper_id=EilbeckEnolskiiPreviato2001, arXiv id=none located, theorem_id=Proposition 3.1 and the paragraph after equation (17), prove: commuting coprime-order ordinary differential operators whose lower coefficients are meromorphic doubly-periodic functions have a Burchnall–Chaundy spectral curve covering the elliptic curve of periods, and this cover is tangential in the Treibich–Verdier sense. The integer Lamé operator is finite-gap and its commuting odd-order operator has Maier's spectral curve, so Maier's cover is tangential at infinity.

(3) E. Colombo, G. P. Pirola, and E. Previato, *Density of elliptic solitons*, paper_id=ColomboPirolaPreviato1994, arXiv id=none, theorem_id=Theorem 1.3, prove the complete cotangent criterion: for a finite cover $\pi : (\Gamma, p) \to (E, 0)$, if $W_E = \pi^* H^0(E, K_E)$ and orthogonality is for the intersection polarization, then

$$\pi \text{ is tangential at } p \quad \Longleftrightarrow \quad W_E^{\perp} = H^0(\Gamma, K_\Gamma(-p)).$$

(4) Sara Arias-de-Reyna, Cécile Armana, Valentijn Karemaker, Marusia Rebolledo, Lara Thomas, and Núria Vila, *Large Galois images for Jacobian varieties of genus 3 curves*, paper_id=AriasEtAl2016, arXiv id=1507.05913, theorem_id=Definition 3.2 and Proposition 3.4, prove: a simple ordinary abelian threefold over an odd finite field $\mathbf{F}_q$ is absolutely simple unless its Frobenius polynomial is $T^6 + cT^3 + q^3$, with $(c, q) = 1$; in the exceptional case it splits over $\mathbf{F}_{q^3}$. For a Weil polynomial $T^6 + aT^5 + bT^4 + cT^3 + qbT^2 + q^2 aT + q^3$, ordinarity is equivalent to $q \nmid c$.

We now specialize the Lamé construction to

$$g_2 = 4, \qquad g_3 = 1, \qquad E_0 : y^2 = 4x^3 - 4x - 1.$$

Its discriminant is $37 \neq 0$, and

$$\mathsf{L}_4 = (B^3 - 208B + 560) \prod_{4e^3-4e-1=0} (B^2 + 10eB - 35e^2 - 28).$$

The product and Maier's symmetric cover formulas are rational; clearing the square denominator by $Y = 4\nu$ gives an integral model. Modulo 11, its monic branch polynomial is

$$\bar{L}(B) = B^9 + B^4 + 6B^3 + 9B^2 + 7B + 5$$

and has the factorization

$$\bar{L} = (B+2)(B^2+9B+5)(B^3+5B^2+6B+1)(B^3+6B^2+7B+6).$$

Thus $\gcd(\bar{L}, \bar{L}') = 1$, so Γ_0 is smooth of genus 4 in characteristics 0 and 11. The first two cited results give a degree-ten tangential cover

$$\pi_0 : (\Gamma_0, p_0) \longrightarrow (E_0, 0),$$

where p_0 is the unique point over $B = \infty$.

For an exact finite-field certificate, evaluate

$$N_r = 11^r + 1 + \sum_{b \in \mathbf{F}_{11^r}} \chi_r\big(\bar{L}(b)\big),$$

where $\chi_r(0) = 0$ and $\chi_r(z) = z^{(11^r-1)/2}$ otherwise. Direct enumeration gives

$$(N_1, N_2, N_3, N_4) = (10, 120, 1312, 15312).$$

For reproducibility, the fields for $r = 2, 3, 4$ are defined respectively by $z^2 + 1$, $z^3 + 4z^2 + 1$, and $z^4 + 4z^3 + 1$, each irreducible by the Frobenius-gcd test; the corresponding character sums are $-2, -20, 670$ (and -2 for $r = 1$). Newton identities and the functional equation give

$$P_{\Gamma_0,11}(T) = T^8 - 2T^7 + T^6 - 6T^5 + 180T^4 - 66T^3 + 121T^2 - 2662T + 14641.$$

Also $\#E_0(\mathbf{F}_{11}) = 7$, hence

$$P_{E_0,11}(T) = T^2 - 5T + 11.$$

Good reduction and $\pi_{0,*}\pi_0^* = [10]$ extend the elliptic factor to the special fiber; it cannot collapse since $11 \nmid 10$. The complementary threefold therefore has polynomial

$$\begin{aligned} h(T) &= P_{\Gamma_0,11}(T) / P_{E_0,11}(T) \\ &= T^6 + 3T^5 + 5T^4 - 14T^3 + 55T^2 + 363T + 1331. \end{aligned}$$

Modulo 3,

$$\bar{h} = T^6 - T^4 + T^3 + T^2 - 1.$$

Modular exponentiation gives

$$T^9 - T \equiv 2 + T^2 + T^3 + T^4, \qquad T^{27} - T \equiv T + 2T^2 + T^3 + 2T^5 \pmod{\bar{h}},$$

the first two remainders have gcd 1 with $\bar{h}$, and $T^{729} - T \equiv 0 \pmod{\bar{h}}$. The degree-six Frobenius test makes h irreducible, hence $A_{0,11}$ simple. It is ordinary since $11 \nmid -14$, and its polynomial is not $T^6 + cT^3 + 11^3$; Arias et al. therefore make it absolutely simple. Absolute simplicity lifts: after finite base extension, the closure

of any proper positive-dimensional geometric abelian subvariety in characteristic zero would give such an abelian subvariety in this good special fiber.

Finally, tangentiality and Theorem 1.3 give

$$\big(\pi_0^* H^0(E_0, K_{E_0})\big)^{\perp} = H^0(\Gamma_0, K_{\Gamma_0}(-p_0)).$$

The left side is precisely the cotangent space of the complementary quotient A_0, proving the last assertion. □

Lemma 2. *There is a smooth nonhyperelliptic genus-four curve $X/\mathbf{C}$, an even theta characteristic L with*

$$L^2 \simeq K_X, \qquad h^0(X, L) = 2,$$

and a simple abelian threefold quotient $q : JX \twoheadrightarrow A$ such that

$$q^* H^0(A, \Omega_A^1) = \operatorname{im}\big(\operatorname{Sym}^2 H^0(X, L) \longrightarrow H^0(X, K_X)\big)\,.$$

Proof. For the seed put $L_0 = \mathcal{O}_{\Gamma_0}(3p_0)$. The hyperelliptic gap sequence gives $K_{\Gamma_0} = \mathcal{O}(6p_0)$ and $h^0(L_0) = 2$. Thus L_0 is an even theta characteristic, and

$$\operatorname{Sym}^2 H^0(L_0) = H^0(K_{\Gamma_0}(-p_0));$$

both spaces have dimension three, and in the odd-degree model the three products are $1, B, B^2$, vanishing at p_0 to orders $6, 4, 2$ as canonical forms.

Take a small marked spin-deformation space. It has dimension 9. Its theta-null locus $\mathcal{S} = \{h^0(L) \geq 2\}$ is one theta-constant equation, so every component has dimension at least 8. Near the seed, upper semicontinuity and even parity give $h^0(L) = 2$ on $\mathcal{S}$; hence $\mathcal{W} = \operatorname{Sym}^2 H^0(L)$ is a holomorphic rank-three subbundle of the Hodge bundle.

The marking makes $P_{\mathbf{Q}} = H^1(A_0, \mathbf{Q})$ a fixed rational six-plane in rank-eight cohomology. The aligned locus $\mathcal{Z} = \{\mathcal{W} \subset P_{\mathbf{C}}\}$ is the zero locus of $\mathcal{W} \to \mathcal{H}_{\mathbf{C}}/P_{\mathbf{C}}$, a map of ranks $3 \to 2$. Krull's height bound gives $\dim Z \geq 8 - 6 = 2$ for every component Z through the seed. On Z, reality of P gives $\overline{\mathcal{W}} \subset P_{\mathbf{C}}$; type considerations and dimensions then give

$$P_{\mathbf{C}} = \mathcal{W} \oplus \overline{\mathcal{W}}.$$

Thus $P_{\mathbf{Q}}$ remains a Hodge substructure with $F^1 P = \mathcal{W}$.

We use one more external result. Armando Treibich, *Hyperelliptic tangential covers and even elliptic finite-gap potentials, back and forth*, paper_id=Treibich2025, arXiv id=2501.16483, theorem_id=Propositions 2.3(3)–(4) and 2.5(3), proves this complete statement. For fixed elliptic E, type μ, and $d \geq 0$, set $n = (\sum_i \mu_i^2 - 1 + 4d)/2$. The cover–Severi correspondence identifies degree-n hyperelliptic tangential covers of type (μ, d) with the rational irreducible members of $|\gamma_E(\mu, d)|$, and Proposition 2.5(3) says that this Severi set is finite (possibly empty).

On the local hyperelliptic branch, theta-characteristic classification gives $L = \mathcal{O}_X(3p)$ for a marked Weierstrass point, hence $\mathcal{W} = H^0(K_X(-p))$. Put $Q = P^{\perp}$. The fixed rational projector e_Q is Hodge along Z; after one fixed integer clears its denominators, it is induced by an abelian homomorphism with elliptic image E_z. Composing with Abel–Jacobi gives $X_z \to E_z$, of locally constant degree because the marked integral homology map is fixed. By the Colombo–Pirola–Previato criterion in Lemma 1, this is a hyperelliptic tangential cover at p. For fixed degree, the

relation $2n+1=\sum_i \mu_i^2+4d$ permits only finitely many (μ,d); Treibich therefore gives finite fibers over each E_z. Elliptic curves have one modulus, so

$$\dim_{\mathbf{C}}(Z\cap\mathcal{H}_4)\le 1.$$

Since $\dim Z\ge 2$, the component Z contains nonhyperelliptic points.

For every nonzero proper rational $R\subset P_{\mathbf{Q}}$, the condition that R be Hodge is closed analytic and proper on Z, since it fails at the simple seed. There are countably many R. Baire therefore supplies a point of

$$Z\setminus\left(\mathcal{H}_4\cup\bigcup_{0\neq R\subsetneq P_{\mathbf{Q}}}\{R\text{ is Hodge}\}\right).$$

The resulting (X,L) is nonhyperelliptic and P is simple. The equivalence between polarizable weight-one rational Hodge structures and complex abelian varieties up to isogeny turns $P\subset H^1(X)$ into a simple threefold quotient $q:JX\twoheadrightarrow A$, with pulled-back cotangent space $\operatorname{Sym}^2 H^0(L)$. □

Lemma 3. *Let A be a simple complex abelian variety of dimension greater than one. If $u\in\operatorname{Aut}_{\mathrm{gp}}(A)$ satisfies*

$$du_0=\lambda\,\mathrm{id}_{T_0A},$$

then $\lambda=1$ and $u=\mathrm{id}$, or $\lambda=-1$ and $u=[-1]$.

Proof. On $H^1(A,\mathbf{C})=H^{1,0}\oplus H^{0,1}$, the only eigenvalues of the rational operator u^* are λ and $\bar\lambda$. Thus the minimal polynomial of λ over $\mathbf{Q}$ has degree at most two.

Suppose first that $\lambda\notin\mathbf{R}$. Then $K=\mathbf{Q}(\lambda)$ is imaginary quadratic, and $H^1(A,\mathbf{Q})$ is a K-vector space of dimension $\dim A>1$. Write $\lambda=a+ib$, with $b\neq 0$. On the underlying real Hodge structure the complex structure is

$$J=\frac{u^*-a\,\mathrm{id}}{b}.$$

Consequently the realification of every one-dimensional K-subspace of $H^1(A,\mathbf{Q})$ is J-stable. It is a two-dimensional rational Hodge substructure and, by the polarization, yields an elliptic factor of A. This contradicts simplicity.

Thus $\lambda\in\mathbf{R}$. The rational operator u^* then has the sole eigenvalue λ, so its trace shows $\lambda\in\mathbf{Q}$. Since both u^* and its inverse preserve the integral lattice, λ is a rational algebraic unit, hence $\lambda=\pm1$. Finally a homomorphism of complex abelian varieties is determined by its differential: a homomorphism with zero differential has connected image of dimension zero and is therefore zero. Hence $u=[\lambda]$. □

Lemma 4. *For the pair (X,L) and quotient $q:JX\twoheadrightarrow A$ supplied by Lemma 2, let $f:X\to A$ be an Abel–Jacobi map followed by q, and put $C=f(X)$. Then:*

1. *A is a simple abelian threefold and C is an irreducible generating curve;*
2. *$f:X\to C$ is the normalization map and is immersive;*
3. *the tangent Gauss map of C has generic degree 3;*
4. *no affine automorphism $t_a\circ u$ satisfying $du_0=\lambda\,\mathrm{id}$ preserves C nontrivially.*

Proof. Write $H^0(X, L) = \langle s, t\rangle$. Since $\deg L = 3$, $h^0(L) = 2$, and X is nonhyperelliptic, L is base-point-free: a base point would leave a degree-two pencil and make X hyperelliptic. Therefore

$$\varphi_L = [s : t] : X \longrightarrow \mathbf{P}^1$$

has degree three. The pullbacks of invariant one-forms on A are

$$W = \operatorname{Sym}^2 H^0(L) = \langle s^2, st, t^2\rangle.$$

They have no common zero, so f is immersive. Since the Abel curve generates JX and q is surjective, C generates A.

Identifying $\mathbf{P}(T_0A) = \mathbf{P}(W^\vee)$, evaluation of these three forms shows that the tangent Gauss map pulled back to X is

$$\gamma = [s^2 : st : t^2] = \nu_2 \circ \varphi_L,$$

where $\nu_2 : \mathbf{P}^1 \hookrightarrow \mathbf{P}^2$ is the quadratic Veronese map. Hence $\deg \gamma = 3$.

Let Y be the normalization of C, and suppose the induced map $r : X \to Y$ has degree $e > 1$. Every form in W comes from Y, so γ factors through r; consequently $e \mid 3$. Thus $e = 3$, and the induced map $Y \to \nu_2(\mathbf{P}^1)$ has degree one. This makes $Y \simeq \mathbf{P}^1$, impossible because an abelian variety contains no nonconstant rational curve. Therefore $e = 1$, so X is the normalization of C. In particular the generic degree of γ_C is three.

It remains to exclude the exceptional symmetry. Suppose $\tau = t_a \circ u$ preserves C and acts nontrivially on it. It lifts uniquely to an automorphism σ of the normalization X. Since $d\tau$ is scalar, it preserves every tangent line, and hence

$$\gamma \circ \sigma = \gamma.$$

By Lemma 3, $u = [1]$ or $[-1]$.

If $u = [-1]$, then $t_a \circ [-1]$ is an involution. Its nontrivial lift σ would be an order-two automorphism over the degree-three map γ. The map would factor through $X/\langle\sigma\rangle$, forcing $2 \mid 3$, a contradiction.

If $u = [1]$, then $\tau = t_a$. Here $a \neq 0$, and the lift σ has finite order because $g(X) = 4$. Since σ acts on each generic three-point fiber of γ, its order divides three; nontriviality forces order three, and γ is the quotient map followed by an isomorphism

$$X/\langle\sigma\rangle \simeq \mathbf{P}^1.$$

Indeed, the cyclic group generated by σ acts freely on the generic fiber, so its order divides the cardinality three. Also, if m is the order of σ, then t_{ma} fixes every point of C, whence $ma = 0$. A nonzero translation has no fixed point, so σ acts freely on X. Riemann–Hurwitz for a free order-three action would give

$$2g(X) - 2 = 3\big(2g(X/\langle\sigma\rangle) - 2\big),$$

which reads $6 = -6$ when the quotient is $\mathbf{P}^1$. This is impossible. Thus no such affine symmetry exists. □

Theorem 5. *Let A be a simple abelian variety of dimension $n > 2$ over an algebraically closed field of characteristic 0, and let $C \subset A$ be an irreducible curve not contained in a translate of any proper abelian subvariety of A. Define*

$$\gamma_C : C^{\mathrm{sm}} \longrightarrow \mathbf{P}(T_0A), \qquad p \longmapsto d(t_{-p})(T_pC).$$

Is γ_C generically injective unless there exists an automorphism

$$\tau = t_a \circ u \in \mathrm{Aut}(A)$$

such that $\tau(C) = C$, $\tau|_C$ is nontrivial on a dense open subset, and

$$du_0 = \lambda \,\mathrm{id}_{T_0 A}$$

for some scalar λ?

Proof. No. It is enough to give a counterexample over the algebraically closed characteristic-zero field $\mathbf{C}$. Lemma 2 supplies a simple abelian threefold A, a genus-four spin curve (X, L), and a quotient $JX \twoheadrightarrow A$. For its image $C \subset A$, Lemma 4 proves that C is irreducible and generates A, so it is not contained in a translate of a proper abelian subvariety. Its tangent Gauss map has generic degree 3, and hence is not generically injective. The same lemma proves that no affine automorphism with scalar linear differential preserves C nontrivially. Thus (A, C) violates the proposed alternative. □